

Normal mean shift and options

For $X \sim N(0,1)$, choose $Y \sim N(\theta,1)$. The likelihood ratio is

$$\frac{f(x)}{g(x)} = \exp\left(-\theta x + \frac{1}{2}\theta^2\right).$$

Mode matching sets $\theta = \arg\max_x h(x)f(x)$. For a binary call,

$$\mathbb{E}[e^{-rT} \mathbf{1}_{\{S_T \geq K\}}] = \mathbb{E}[e^{-rT} \mathbf{1}_{\{X \geq b\}}],$$

where

$$b = \frac{\log(K/S_0) - (r - \sigma^2/2)T}{\sigma\sqrt{T}}.$$

Here $h(x) = e^{-rT} \mathbf{1}_{\{x \geq b\}}$, $x^* = \max\{b,0\}$, and with $Y \sim N(x^*,1)$ the output is

$$h(Y) = \frac{f(Y)}{g(Y)} = \begin{cases} 0, & Y < b, \\ \exp\{-rT - x^*Y + (x^*)^2/2\}, & Y \geq b. \end{cases}$$

For a vanilla call,

$$h(x) = \left(S_0 \exp\left\{-\frac{1}{2}\sigma^2 T + \sigma\sqrt{T}x\right\} - e^{-rT}K\right)^+,$$

and the shift pushes sampling toward exercise states.

Exponential tilting

For density f , the exponential tilt family is

$$f_\theta(x) = \frac{e^{\theta x f(x)}}{\mathbb{E}[e^{\theta X}]}, \qquad \theta \in \mathbb{R}.$$

For a pmf p ,

$$p_\theta(x) = \frac{e^{\theta x p(x)}}{\mathbb{E}[e^{\theta X}]},$$

For normal X , this is mean shifting. For rare large losses, take $\theta > 0$.

Bernoulli rare event

Let independent defaults satisfy $P(X_i = 1) = p_k$ and $L = \sum_{k=1}^m X_k$. To estimate

$P(L > x)$ with $x > \mathbb{E}[L]$, tilt Bernoulli:

$$\bar{p}_k = P(Y_k = 1) = \frac{p_k e^\theta}{1 + p_k(e^\theta - 1)}, \qquad \bar{L} = \sum_{k=1}^m Y_k.$$

The IS output is

$$H = \mathbf{1}_{\{L > x\}} \prod_{k=1}^m \left(\frac{p_k}{\bar{p}_k}\right)^{Y_k} \left(\frac{1 - p_k}{1 - \bar{p}_k}\right)^{1 - Y_k}.$$

With

$$\theta(\theta) = \sum_{k=1}^m \log[1 + p_k(e^\theta - 1)],$$

one has $E[H^2] \leq e^{-2\theta x + 2\theta(\theta)}$. Choose $\theta > 0$ by solving $\theta'(\theta) = x$.

4.5 Cross-Entropy Method

Principle

Cross-entropy selects a parametric IS density f_θ close to the zero-variance density

$$g^*(x) = \tfrac{1}{h} h(x) f(x).$$

Minimize

$$J(g^*|f_\theta) = \int \log\left(\frac{g^*(x)}{f_\theta(x)}\right) g^*(x) dx,$$

equivalently maximize

$$\int h(x) f(x) \log f_\theta(x) dx.$$

The population first-order equation is

$$\mathbb{E}\left[h(X) \frac{\partial}{\partial \theta} \log f_\theta(X)\right] = 0.$$

Basic algorithm

Using pilot samples $X_1, \dots, X_N \sim f$, solve

$$\frac{1}{N} \sum_{k=1}^N h(X_k) \frac{\partial}{\partial \theta} \log f_\theta(X_k) = 0.$$

For $f = N(0, I_m)$ and $f_\theta = N(\theta, I_m)$,

$$\theta = \frac{\sum_{k=1}^N h(X_k) X_k}{\sum_{k=1}^N h(X_k)}.$$

Then sample $Y_i \sim f_\theta$ and average

$$H_i = h(Y_i) \frac{f(Y_i)}{f_\theta(Y_i)}.$$

For a one-dimensional normal shift, $H_i = h(Y_i) \exp(-\hat{\theta} Y_i + \hat{\theta}^2/2)$. For very rare events, the basic method may fail because $\sum_k h(X_k) = 0$ in the pilot run.

Iterative cross-entropy

Let $g(x) = f(x)/f_\theta(x)$. If pilot samples are drawn from f_V , then maximize

$$\mathbb{E}_V[h(Y) \ell_V(Y) \log f_\theta(Y)],$$

and solve

$$\mathbb{E}_V\left[h(Y) \ell_V(Y) \frac{\partial}{\partial \theta} \log f_\theta(Y)\right] = 0.$$

At iteration j , sample $Y_k \sim f_{\hat{\theta}_j}$ and update from

$$\frac{1}{N} \sum_{k=1}^N h(Y_k) \ell_{\hat{\theta}_j}(Y_k) \frac{\partial}{\partial \theta} \log f_\theta(Y_k) = 0.$$

For $N(\theta, I_m)$,

$$\hat{\theta}^{j+1} = \frac{\sum_{k=1}^N h(Y_k) \ell_{\hat{\theta}_j}(Y_k) Y_k}{\sum_{k=1}^N h(Y_k) \ell_{\hat{\theta}_j}(Y_k)}.$$

Here

$$\ell_{\hat{\theta}_j}(Y_k) = \exp(-\langle \hat{\theta}^j, Y_k \rangle + \|\hat{\theta}^j\|^2/2).$$

Usually only a few iterations are needed; convergence means the updates oscillate in a small range rather than become deterministic.

5 Discretization Methods

5.1 Euler And Milstein Schemes

Euler scheme

For the SDE $dX(t) = a(X(t))dt + b(X(t))dW(t)$, with $X \in \mathbb{R}^d$ and $W \in \mathbb{R}^m$, the Euler approximation on $t_i = ih$ is

$$\hat{X}(i+1) = \hat{X}(i) + a(\hat{X}(i))h + b(\hat{X}(i))\sqrt{\Delta t}Z_{i+1},$$

where Z_i are independent m -dimensional standard normals. Euler freezes the drift and diffusion integrands on each interval. The drift is order h , while the diffusion increment is order $\sqrt{h}$, so a first refinement focuses on the diffusion term.

Scalar Milstein refinement

In the scalar case, Ito's formula gives

$$db(X(t)) = \left[b'(X(t))a(X(t)) + \tfrac{1}{2}b''(X(t))b^2(X(t))\right]dt + b'(X(t))b(X(t))dW(t).$$

Dropping the higher-order drift part,

$$b(X(u)) \approx b(X(t)) + b'(X(t))b(X(t))[W(u) - W(t)].$$

With $\Delta W = W(t+h) - W(t)$,

$$f_t^{t+h}b(X(u))dW(u) \approx b(X(t))\Delta W + b'(X(t))b(X(t))\int_t^{t+h} [W(u) - W(t)]dW(u).$$

The remaining integral is

$$f_t^{t+h}[W(u) - W(t)]dW(u) = \tfrac{1}{2}(\Delta W)^2 - \tfrac{1}{2}h.$$

Thus the Milstein one-step approximation is $X(t+h) \approx X(t) + a(X(t))h + b(X(t))\Delta W + \tfrac{1}{2}b'(X(t))b(X(t))\{(\Delta W)^2 - h\}$.

Equivalently, with $\Delta W = \sqrt{h}Z_{t+1}$,

$$\hat{X}(i+1) = \hat{X}(i) + a(\hat{X}(i))h + b(\hat{X}(i))\sqrt{h}Z_{i+1} + \tfrac{1}{2}b'(\hat{X}(i))b(\hat{X}(i))h(Z_{i+1}^2 - 1).$$

Conditional on $\hat{X}(i)$, the extra term has mean 0 and is uncorrelated with the Euler diffusion term since $Z^2 - 1$ and Z are uncorrelated.

GBM comparison

For $dX_t = rX_t dt + \sigma X_t dW_t$, Euler is

$$\hat{X}_{t+1} = \hat{X}_t + r\hat{X}_t h + \sigma \hat{X}_t \sqrt{h}Z_{t+1}.$$

Milstein is

$$\hat{X}_{t+1} = \hat{X}_t + r\hat{X}_t h + \sigma \hat{X}_t \sqrt{h}Z_{t+1} + \tfrac{1}{2}\sigma^2 \hat{X}_t h(Z_{t+1}^2 - 1).$$

The exact benchmark is

$$X_{t+1} = X_t \exp\left((r - \tfrac{1}{2}\sigma^2)h + \sigma\sqrt{h}Z_{t+1}\right).$$

In the lecture's numerical comparison, Euler strong error is order $\sqrt{h}$, while Milstein error is order h .

Multidimensional difficulty

For components X_i, W_j, a_i , and b_{ij} ,

$$X_i(t+h) = X_i(t) + f_t^{t+h}a_i(X(u))du + \sum_{j=1}^m \int_t^{t+h} b_{ij}(X(u))dW_j(u).$$

The diffusion refinement contains

$$f_t^{t+h}b_{ij}(X(u))dW_j(u) \approx b_{ij}(X(t))\Delta W_j$$

$$+ \sum_{j'=1}^m \sum_{k=1}^m \frac{\partial b_{ij}}{\partial x_{k'}}(X(t))\rho_{ik}(X(t))\Delta k_j,$$

where

$$I_{kj} = \int_t^{t+h} [W_k(u) - W_k(t)]dW_j(u).$$

For $k = j$, $I_{jj} = \tfrac{1}{2}(\Delta W_j)^2 - \tfrac{1}{2}h$. For $k \neq j$, the mixed integrals have no comparable simple formula; their differences are Levy area terms, and sampling them is difficult.

5.2 Lamperti Transform

Definition

For $dX_t = b(X_t)dt + \sigma(X_t)dW_t$ with $\sigma(x) > 0$, define

$$Y_t = F(X_t), \qquad F(x) = \int \frac{1}{\sigma(x)} dx.$$

Then F is increasing and $X_t = F^{-1}(Y_t)$. Since

$$F'(x) = 1/\sigma(x), \qquad F''(x) = -\sigma'(x)/\sigma^2(x),$$

Ito's formula gives

$$dY_t = \left[\frac{b(X_t)}{\sigma(X_t)} - \tfrac{1}{2}\sigma'(X_t)\right]dt + dW_t.$$

Writing $X_t = F^{-1}(Y_t)$,

$$dY_t = a(Y_t)dt + dW_t, \qquad a(y) = \frac{b(F^{-1}(y))}{\sigma(F^{-1}(y))} - \tfrac{1}{2}\sigma'(F^{-1}(y)).$$

Thus Lamperti converts state-dependent volatility into constant volatility one. Euler through Lamperti uses

$$\hat{Y}_{t+1} = \hat{Y}_t + a(\hat{Y}_t)\Delta t_{t+1} + \sigma\sqrt{\Delta t_{t+1}}Z_{t+1}, \qquad \hat{X}_{t+1} = F^{-1}(\hat{Y}_{t+1}).$$

GBM transform

For $dX_t = rX_t dt + \sigma X_t dW_t$,

$$F(x) = \tfrac{1}{\sigma} \log x, \qquad Y_t = \tfrac{1}{\sigma} \log X_t, \qquad X_t = e^{\sigma Y_t}.$$

Ito's formula gives

$$dY_t = \left(\tfrac{r}{\sigma} - \tfrac{\sigma}{2}\right)dt + dW_t.$$

Euler on Y is exact, so transforming back gives a scheme for X without discretization error.

5.3 Numerical Examples

Exact grid for deterministic volatility

For a discretely monitored floating-strike lookback call with payoff $S_T - \min_{i=1,\dots,m} S_i$ and $dS_t = rS_t dt + \theta(r)S_t dW_t$, set $Y_t = \log S_t$. Then $dY_t = (r - \tfrac{1}{2}\theta^2(t))dt + \theta(r)dW_t$.

Let $\sigma_t^2 = \int_t^{t+1} \theta^2(r)dt$. The exact grid recursion is

$$Y_{i+1} = Y_i + rY_i \Delta t_i - \tfrac{1}{2}\sigma_t^2 + \sigma_i Z_{i+1}, \qquad S_{i+1} = e^{Y_{i+1}}.$$

The discounted output is $H = e^{-rT}(S_m - \min(S_1, \dots, S_m))$. For $\theta(t) = \sqrt{a^2 + b^2 \sin(2\pi t/T)}$,

$$\sigma_t^2 = a^2 \Delta t_i + \frac{b^2 T}{4\pi^2} [\cos(2\pi t_i/T) - \cos(2\pi i+1/T)].$$

Exact grid simulation removes process/discretization error, but a continuously monitored payoff can still be biased because the between-grid minimum is approximated.

Artificial-dynamics control

For $dS_t = rS_t dt + \theta_t S_t dW_t$ and $d\theta_t = a(\Theta - \theta_t)dt + \beta dB_t$, with $\text{corr}(dW, dB) = \rho$, set $Y_t = \log S_t$. Euler gives

$$\hat{Y}_{t+1} = \hat{Y}_t + (r - \tfrac{1}{2}\hat{\theta}_t^2)\Delta t_i + \hat{\theta}_t \sqrt{\Delta t_i}Z_{t+1},$$

$$\hat{\theta}_{t+1} = \hat{\theta}_t + a(\Theta - \hat{\theta}_t)\Delta t_i + \beta \sqrt{\Delta t_i}R_{t+1}.$$

The call output is $H = e^{-rT}(e^{Y_T} - K)^+$. Use the same Z_t to define an artificial GBM log process

$$\bar{Y}_{i+1} = \bar{Y}_i + (r - \tfrac{1}{2}\sigma^2)\Delta t_i + \sigma\sqrt{\Delta t_i}Z_{i+1}.$$

Then $V = e^{-rT}(e^{\bar{Y}_T} - K)^+$ has expectation BSCall(S_0, K, T, r, σ), giving the control output

$X - b[V - \text{BSCall}(S_0, K, T, r, \sigma)]$. Choosing σ near the long-run mean Θ gives strong variance reduction in the lecture example.

Heston with cross-entropy IS

For the Heston model

$$dS_t = rS_t dt + \sqrt{\theta_t}S_t dW_t, \qquad d\theta_t = a(b - \theta_t)dt + \sigma\sqrt{\theta_t}dB_t,$$

use the nonnegative Euler scheme

$$\hat{S}_{t+1} = \hat{S}_t + r\hat{S}_t \Delta t_i + \hat{S}_t \sqrt{\hat{\theta}_t} \Delta t_i Z_{i+1},$$

$$\hat{\theta}_{t+1} = \max\{0, \hat{\theta}_t + a(b - \hat{\theta}_t)\Delta t_i + \sigma\sqrt{\hat{\theta}_t} \Delta t_i R_{i+1}\},$$

where $R_{i+1} = \rho Z_{i+1} + \sqrt{1 - \rho^2}Y_{i+1}$. The payoff is $h(X) = e^{-rT}(S_T - K)^+$ for $X = (Z_1, \dots, Z_m, Y_1, \dots, Y_m)$.

For IS use $N(\mu, I_{2m})$, $\mu = (v, v)$. An initial tilt is

$$v^0 = x(\sqrt{\Delta t_1}, \dots, \sqrt{\Delta t_m}), \qquad v^0 = -\frac{\rho}{1 - \rho^2}v^0,$$

where matching $E[S_T] \approx K$ after replacing θ_t by b gives

$$x = \frac{1}{\sqrt{\Delta t_1}} \log\left(\frac{K}{S_0}\right) - \frac{\rho}{\sqrt{\Delta t_1}}.$$

The cross-entropy update is

$$\hat{\mu}^{i+1} = \frac{\sum_{k=1}^N \mathbf{1}_{h(\bar{X}_k) \leq -(\mu^i)^T \bar{X}_k}}{\sum_{k=1}^N h(\bar{X}_k)e^{-(\mu^i)^T \bar{X}_k}} \bar{X}_k, \qquad \bar{X}_k \sim N(\mu^i, I_{2m}).$$

After $\hat{\mu} = (\hat{v}, \hat{v})$ is chosen, simulate $Z_t \sim N(\hat{v}_1, 1)$ and $Y_t \sim N(\hat{v}_1, 1)$ and multiply the payoff by

$$\exp\left\{-\sum_{i=1}^m (\hat{v}_i Z_i + \hat{v}_i Y_i) + \tfrac{1}{2}\sum_{i=1}^m (\hat{v}_i^2 + \hat{v}_i^2)\right\}.$$

The variance reduction is large for high strikes.

6 Estimating Sensitivities

6.1 Finite Differences

Bias

Let $\alpha(\theta) = \mathbb{E}[Y(\theta)]$. The forward-difference estimator is

$$\hat{\Delta}_F(n, h) = \frac{\bar{Y}_n(\theta + h) - \bar{Y}_n(\theta)}{h}.$$

Since $\mathbb{E}[\hat{\Delta}_F] = h^{-1}(\alpha(\theta + h) - \alpha(\theta))$, if α is twice differentiable,

$$\text{Bias}(\hat{\Delta}_F) = \mathbb{E}[\hat{\Delta}_F - \alpha'(\theta)] = \tfrac{1}{2}\alpha''(\theta)h + o(h).$$

The central-difference estimator is

$$\hat{\Delta}_C(n, h) = \frac{\bar{Y}_n(\theta + h) - \bar{Y}_n(\theta - h)}{2h}.$$

Taylor expansion gives $\text{Bias}(\hat{\Delta}_C) = o(h)$; if $\alpha'''(\theta)$ exists,

$$\text{Bias}(\hat{\Delta}_C) = \tfrac{1}{6}\alpha'''(\theta)h^2 + o(h^2).$$

Central difference losses more if $\alpha(\theta)$ is also needed, but has smaller bias order.

Variance and common random numbers

$$\text{var}[\hat{\Delta}_F] = h^{-2} \text{var}[\bar{Y}_n(\theta + h) - \bar{Y}_n(\theta)].$$

$$\text{var}[\hat{\Delta}_C] = (2h)^{-2} \text{var}[\bar{Y}_n(\theta + h) - \bar{Y}_n(\theta - h)].$$

The factor h^{-2} makes very small h dangerous unless n is increased or the two simulations are strongly coupled. For a GBM call delta, use the same Z_t for $S_0 + h$ and $S_0 - h$:

$$X_t = (S_0 + h)e^{(r - \sigma^2/2)T/\sigma\sqrt{\Delta t_i}}, \qquad Y_t = (S_0 - h)e^{(r - \sigma^2/2)T/\sigma\sqrt{\Delta t_i}},$$

$$H_t = \frac{e^{-rT}(X_t - K)^+ - e^{-rT}(Y_t - K)^+}{2h}, \qquad \hat{v} = \tfrac{1}{n} \sum_{i=1}^n H_i.$$

$$\text{SE} = \left[\frac{1}{n(n-1)}\left(\sum_{i=1}^n H_i^2 - n\hat{v}^2\right)\right]^{1/2}.$$

6.2 Pathwise Derivatives

Method

Differentiate the random output:

$$Y'(\theta) = \lim_{h \rightarrow 0} \frac{Y(\theta + h) - Y(\theta)}{h}.$$

It is unbiased for $\alpha'(\theta)$ if

$$\mathbb{E}\left[\frac{d}{d\theta} Y(\theta)\right] = \frac{d}{d\theta} \mathbb{E}[Y(\theta)].$$

Then estimate $\alpha'(\theta)$ by the sample average of copies of $Y'(\theta)$.

Black-Scholes delta

For $Y = e^{-rT}[(S(T) - K)^+]$ and $S(T) = S(0)e^{(r - \sigma^2/2)T/\sigma\sqrt{T}Z}$,

$$\frac{dY}{dS(0)} = \frac{dY}{dS(T)} \frac{dS(T)}{dS(0)}.$$

Since $d(X - K)^+ / dx = \mathbf{1}_{\{x > K\}}$ except at K and $\mathbb{P}(S(T) = K) = 0$,

$$\frac{dY}{dS(0)} = e^{-rT} \frac{S(T)}{S(0)} \mathbf{1}_{\{S(T) > K\}}.$$

Asian delta and vega

For $Y = e^{-rT}[\bar{S} - K]^+$ and $\bar{S} = m^{-1} \sum_{i=1}^m S(t_i)$,

$$\frac$$

8.2 Variance Reduction

Control variate

Evaluate both L_i and Q_i on each path. A control estimator for $P(L > x)$ is

$$1 - F_i^{CV}(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{\{L_i > x\}} - \hat{\beta} \left(\frac{1}{n} \sum_{i=1}^n \mathbf{1}_{\{Q_i > y\}} - P(Q > y) \right).$$

$P(Q > y)$ is known by transform inversion, and y need not equal x . The method can disappoint in the far tail: L and Q may be highly correlated, but their exceedance indicators may not be, and few exceedances make $\hat{\beta}$ noisy.

Importance sampling

Large Q occurs when $b_j Z_j$ is large or when $\lambda_j Z_j^2$ is large with $\lambda_j > 0$. Define the twisted measure

$$\frac{dP_\theta}{dP} = e^{\theta Q - \psi(\theta)}.$$

Then

$$P(L > x) = \mathbb{E}_\theta \left[e^{-\theta Q + \psi(\theta)} \mathbf{1}_{\{L > x\}} \right].$$

Under P_θ , Z remains normal with diagonal covariance:

$$Z \sim N(\mu(\theta), \Sigma(\theta)), \quad \mu_j(\theta) = \frac{\theta b_j}{1 - 2\lambda_j \theta}, \quad \sigma_j^2(\theta) = \frac{1}{1 - 2\lambda_j \theta}.$$

For $\theta > 0$, the mean shifts in the sign of b_j and variance increases when $\lambda_j > 0$.

IS algorithm

Choose $\theta > 0$ with finite $\psi(\theta)$. For each path, generate $Z \sim N(\mu(\theta), \Sigma(\theta))$, compute Q , set $\Delta S = CZ$, revalue the portfolio to get L , and record $e^{-\theta Q + \psi(\theta)} \mathbf{1}_{\{L > x\}}$.

Average over paths. A practical twisting parameter solves

$$\psi'(\theta_*) = x.$$

Since $\psi'(\theta) = \mathbb{E}_\theta[Q]$, this samples under a distribution with $\mathbb{E}_{\theta_*}[Q] = x$, moving the rare threshold near the center of the new distribution.

Stratified IS

Stratify the IS estimator by Q . For strata $A_1, \dots, A_K$,

$$\mathbb{E}_\theta[e^{-\theta Q + \psi(\theta)} \mathbf{1}_{\{L > x\}}] = \sum_{k=1}^K P_\theta(Q \in A_k) \mathbb{E}_\theta[e^{-\theta Q + \psi(\theta)} \mathbf{1}_{\{L > x\}} \mid Q \in A_k].$$

Under P_θ ,

$$\psi_\theta(u) = \psi(\theta + u) - \psi(\theta), \quad \hat{\phi}_\theta(u) = e^{\psi(\theta + u) - \psi(\theta)}.$$

Use inversion to find $F_\theta(x) = P_\theta(Q \leq x)$ and stratum endpoints a_k with desired probabilities, $A_k = (a_{k-1}, a_k]$.

Conditional sampling uses acceptance-rejection: generate $Z \sim N(\mu(\theta), \Sigma(\theta))$, compute Q , assign to the stratum containing Q , keep it if that stratum still needs samples, otherwise discard. This can be worthwhile because generating Z is cheap relative to revaluation.

With samples (Q_{kj}, L_{kj}) in stratum k , estimate

$$\sum_{k=1}^K P_\theta(Q \in A_k) \frac{1}{n_k} \sum_{j=1}^{n_k} e^{-\theta Q_{kj} + \psi(\theta)} \mathbf{1}_{\{L_{kj} > x\}}.$$

8.3 Numerical Example Notes

The lecture tests short option portfolios on 10 or 100 GBM assets. Prices, deltas, and gammas use Black-Scholes; volatility is 30%, $r = 5\%$, and most cases use uncorrelated assets. Thresholds are set by

$$x = \mathbb{E}[Q] + x_{\text{std}} \sqrt{\text{var}(Q)} = a + \sum_{j=1}^m \lambda_j + x_{\text{std}} \sqrt{\sum_{j=1}^m (b_j^2 + 2\lambda_j^2)}.$$

Reported variance-reduction factors compare ordinary MC with control variates, importance sampling, and importance sampling plus stratification.

9 Markov Chain Monte Carlo Methods

9.1 Bayesian Inference

Prior, likelihood, and posterior

For data $x = (x_1, \dots, x_n)$, parameters $\theta = (\theta_1, \dots, \theta_r)$, likelihood $L(\theta) = f(x \mid \theta)$, and prior $p(\theta)$, Bayes' theorem gives

$$\pi(\theta \mid x) = \frac{f(x \mid \theta)p(\theta)}{f(x)} \propto L(\theta)p(\theta), \quad f(x) = \int \theta p(x \mid \theta)p(\theta) d\theta.$$

A common estimator is the posterior mean $\hat{\theta} = \int \theta \pi(\theta \mid x) d\theta$. A posterior can serve as the prior when new data arrive.

Normal-normal conjugacy

If $x_i \stackrel{iid}{\sim} N(\mu, \sigma^2)$ with known σ^2 and $\mu \sim N(m, \tau_1^2)$, then $\mu \mid x \sim N(m_1, \tau_1^2)$, where

$$m_1 = \frac{\tau^2 \bar{x} + m \sigma^2 / n}{\tau^2 + \sigma^2 / n}, \quad \tau_1^2 = \frac{\tau^2 \sigma^2}{n \tau^2 + \sigma^2}.$$

As $n \rightarrow \infty$, $m_1 \rightarrow \bar{x}$ and $\tau_1^2 \rightarrow 0$. As $\tau^2 \rightarrow \infty$, the prior becomes noninformative and $\mu \mid x \sim N(\bar{x}, \sigma^2 / n)$.

A conjugate prior produces a posterior in the same family, with updated hyperparameters. Common pairs are Poisson–Gamma, Binomial–Beta, normal mean–Normal, and normal variance–Inverse Gamma. The lecture uses

$$f_G(x) = \frac{\beta^\alpha x^{\alpha-1} e^{-\beta x}}{\Gamma(\alpha)}, \quad f_G(x) = \frac{\beta^\alpha x^{\alpha-1} e^{-\beta/x}}{\Gamma(\alpha)}.$$

Importance sampling for posteriors

For an easy density q , draw $\theta_i \sim q$ and estimate

$$\hat{\theta} = \int \theta \frac{\pi(\theta \mid x)}{q(\theta)} d\theta \approx \frac{1}{n} \sum_{i=1}^n \frac{\theta_i \pi(\theta_i \mid x)}{q(\theta_i)}.$$

Here IS facilitates integration rather than primarily reducing variance; poor overlap between q and the posterior may cause large variance.

9.2 Gibbs Sampling

General algorithm

MCMC constructs a Markov chain whose limiting distribution is the target posterior. Gibbs sampling cycles through full conditionals:

$$\theta_j^{(i)} \sim p(\theta_j \mid \theta_1^{(i)}, \dots, \theta_{j-1}^{(i)}, \theta_{j+1}^{(i-1)}, \dots, \theta_r^{(i-1)}, x).$$

Under standard regularity conditions, $\theta^{(i)}$ converges in distribution to $p(\theta \mid x)$. Discard M burn-in draws and estimate

$$\hat{\theta}_j = \frac{1}{n} \sum_{i=1}^n \theta_j^{(M+i)}.$$

Bivariate normal

For $(\theta_1, \theta_2)^T \sim N(0, \Sigma)$ with unit variances and correlation ρ ,

$$\theta_1 \mid \theta_2 \sim N(\rho \theta_2, 1 - \rho^2), \quad \theta_2 \mid \theta_1 \sim N(\rho \theta_1, 1 - \rho^2).$$

Thus, using independent $Z_1, Z_2 \sim N(0, 1)$,

$$\theta_1^{(i)} = \rho \theta_2^{(i-1)} + \sqrt{1 - \rho^2} Z_1, \quad \theta_2^{(i)} = \rho \theta_1^{(i)} + \sqrt{1 - \rho^2} Z_2.$$

Unknown normal mean and variance

For $x_j \sim N(\mu, \sigma^2)$ with both unknown,

$$\mu_i \mid \sigma_{i-1}^2 \sim N(m_i, \tau_i^2), \quad m_i = \frac{\tau_{i-1}^2 \bar{x} + m_{i-1} \sigma_{i-1}^2 / n}{\tau_{i-1}^2 + \sigma_{i-1}^2 / n}, \quad \tau_i^2 = \frac{\tau_{i-1}^2 \sigma_{i-1}^2}{n \tau_{i-1}^2 + \sigma_{i-1}^2}.$$

$$\sigma_i^2 \mid \mu_i \sim IG(\alpha_i, \beta_i), \quad \alpha_i = \alpha_{i-1} + n/2, \quad \beta_i = \beta_{i-1} + \frac{1}{2} \sum_{j=1}^n (x_j - \mu_i)^2.$$

Alternate these draws, discard burn-in, and average the retained values.

9.3 Jump-Diffusion Case Study

Model and latent jumps

Merton's model is $d \log S_t = \mu dt + \sigma dW_t + Y dN_t$, where N_t has intensity λ and $Y \sim N(k, s^2)$. For small Δt ,

$$x_t = \Delta \log S_t = \mu \Delta t + \sigma \Delta W_t + Y_t \Delta N_t, \quad P(\Delta N_t = 1) = \lambda \Delta t.$$

Conditional return laws are

$$x_t \mid \Delta N_t = 0 \sim N(\mu \Delta t, \sigma^2 \Delta t), \quad x_t \mid \Delta N_t = 1 \sim N(\mu \Delta t + k, \sigma^2 \Delta t + s^2).$$

If f_0, f_1 denote these densities, the latent indicator probability is

$$P(\Delta N_t = 1 \mid x_t) = \frac{f_1(x_t) \lambda \Delta t}{f_1(x_t) \lambda \Delta t + f_0(x_t) (1 - \lambda \Delta t)}.$$

When $\Delta N_t = 1$, the latent jump size satisfies

$$Y_t \mid x_t \sim N \left(\frac{(x_t - \mu \Delta t) / (\sigma^2 \Delta t) + k / s^2}{1 / (\sigma^2 \Delta t) + 1 / s^2}, \frac{1}{1 / (\sigma^2 \Delta t) + 1 / s^2} \right).$$

Conditional parameter updates

Given the latent $\Delta N_t, Y_t$, use normal updates for μ, k , inverse-gamma updates for σ^2, s^2 , and a beta update for the jump probability. In particular,

$$\sigma^2 \mid \dots \sim IG \left(\alpha + \frac{n}{2}, \beta + \frac{\sum_i (x_i - \mu \Delta t - Y_i \Delta N_t)^2}{2 \Delta t} \right),$$

$$\lambda \Delta t \mid \dots \sim Be(a + N, b + n - N), \quad N = \sum_i \Delta N_t,$$

$$k \mid \dots \sim N \left(\frac{\tau_0^2 \bar{Y} + m Y s^2 / N}{\tau_0^2 + s^2 / N}, \frac{\tau_0^2 s^2}{N \tau_0^2 + s^2} \right),$$

$$s^2 \mid \dots \sim IG \left(\alpha_Y + \frac{N}{2}, \beta_Y + \frac{1}{2} \sum_{j=1}^N (Y_j - k)^2 \right).$$

One Gibbs sweep updates $\mu, \sigma^2, \lambda, k, s^2$, every ΔN_t , and then Y_t where $\Delta N_t = 1$. The Dow Jones example finds about 1.75–5.5 jumps per year and large jump variances, so jump risk is material.

9.4 Metropolis–Hastings

Detailed balance

For transition probabilities p_{ij} , stationarity requires $\pi(j) = \sum_i \pi(i) p_{ij}$. Detailed balance,

$$\pi(i) p_{ij} = \pi(j) p_{ji},$$

is sufficient. Starting with proposal q_{ij} , accept $i \rightarrow j$ with

$$\alpha_{ij} = \min \left\{ 1, \frac{\pi(j) q_{ji}}{\pi(i) q_{ij}} \right\}.$$

Then $p_{ij} = q_{ij} \alpha_{ij}$ for $i \neq j$, with leftover probability assigned to staying at i ; the resulting chain is reversible with stationary distribution π .

Continuous-state algorithm

At state θ_{k-1} , propose $X_k \sim q(\cdot \mid \theta_{k-1})$ and compute

$$\alpha = \min \left\{ 1, \frac{\pi(X_k) q(\theta_{k-1} \mid X_k)}{\pi(\theta_{k-1}) q(X_k \mid \theta_{k-1})} \right\}.$$

Accept if $U \leq \alpha$; otherwise set $\theta_k = \theta_{k-1}$. Symmetric proposals cancel the q ratio, and unknown target normalizing constants cancel. Gibbs sampling is MH with full-conditional proposals and acceptance probability 1.

9.5 Stochastic Volatility And MH

Let $y_t = e^{h_t/2} \varepsilon_t$ and $h_{t+1} = \mu + \tau \eta_t$, with independent standard normal shocks.

For priors $\mu \sim N(\alpha_\mu, \beta_\mu^2)$ and $\tau^2 \sim IG(\alpha_\tau, \beta_\tau)$,

$$\mu \mid \tau^2, h \sim N(\alpha_\mu, \hat{\beta}_\mu^2), \quad \alpha_\mu = \frac{\bar{h} \beta_\mu^2 + \alpha_\mu \tau^2 / n}{\beta_\mu^2 + \tau^2 / n}, \quad \hat{\beta}_\mu^2 = \frac{\beta_\mu^2 \tau^2}{n \beta_\mu^2 + \tau^2},$$

$$\tau^2 \mid \mu, h \sim IG \left(\alpha_\tau + \frac{n}{2}, \beta_\tau + \frac{1}{2} \sum_{t=1}^n (h_t - \mu)^2 \right).$$

The latent-state conditional is standard:

$$p(h_t \mid \mu, \tau, y, h_{-t}) \propto e^{-h_t/2} \exp \left(-\frac{y_t^2}{2 e^{h_t}} \right) \exp \left(-\frac{(h_t - \mu)^2}{2 \tau^2} \right).$$

Use an independence proposal $x_t \sim q$, e.g. normal, and accept with

$$\min \left\{ 1, \frac{p(x_t \mid \mu, \tau, y, h_{-t}) q(h_t^{(i-1)})}{p(h_t^{(i-1)} \mid \mu, \tau, y, h_{-t}) q(x_t)} \right\}.$$

This is Metropolis-within-Gibbs: use Gibbs for tractable conditionals and MH for nonstandard latent-volatility conditionals.